NLP Project 4

Stress-Testing Pre-Trained Language Models

Sentiment Analysis & QA with BiDAF

Built by Fatulla Bashirov (Task 1 & UI Architecture)
and Kamal Aghzada (Task 2 & BiDAF Pipeline).

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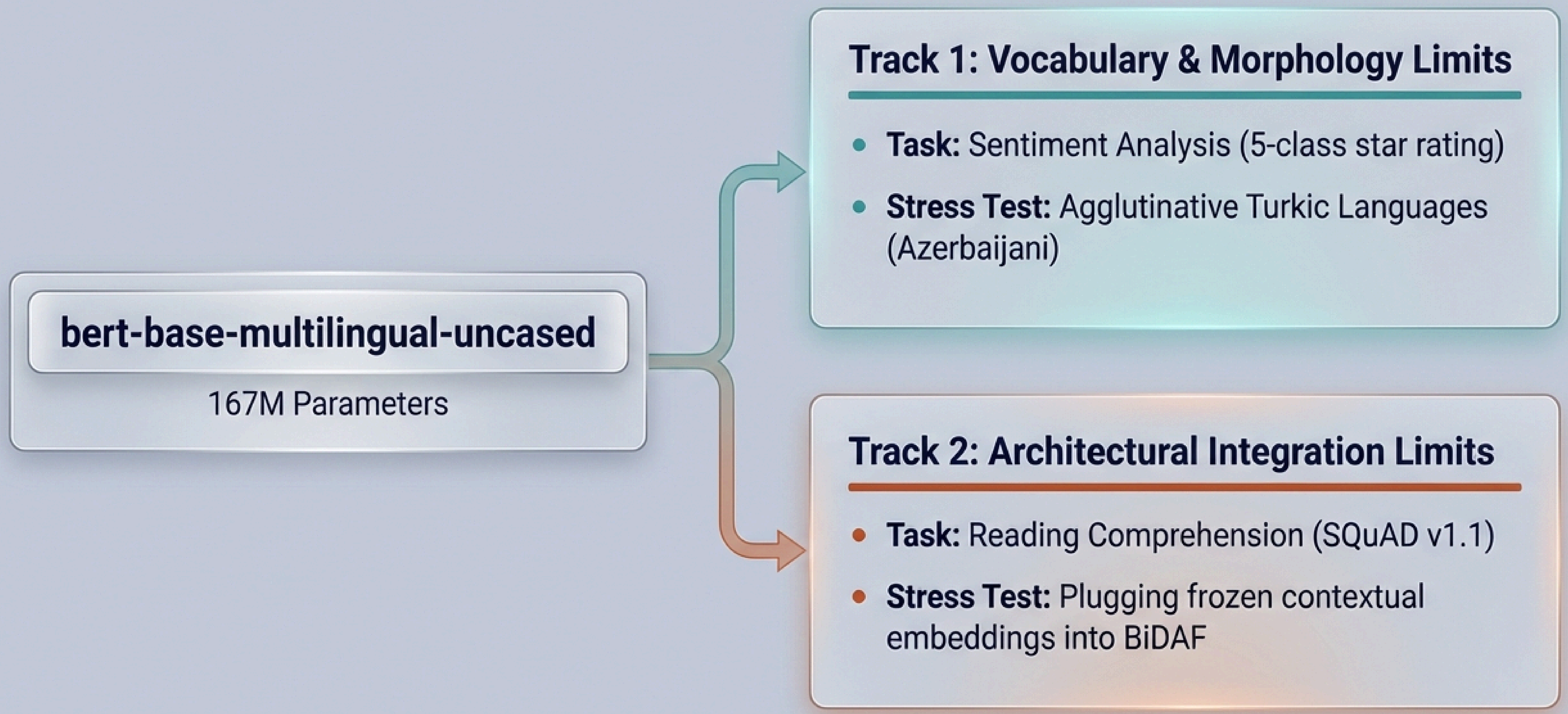

bert-base-multilingual-uncased
167M Parameters
Track 1: Vocabulary & Morphology Limits
Task: Sentiment Analysis (5-class star rating)
Stress Test: Agglutinative Turkic Languages (Azerbaijani)
Track 2: Architectural Integration Limits
Task: Reading Comprehension (SQuAD v1.1)
Stress Test: Plugging frozen contextual embeddings into BiDAF

Track 1: Establishing the Baseline

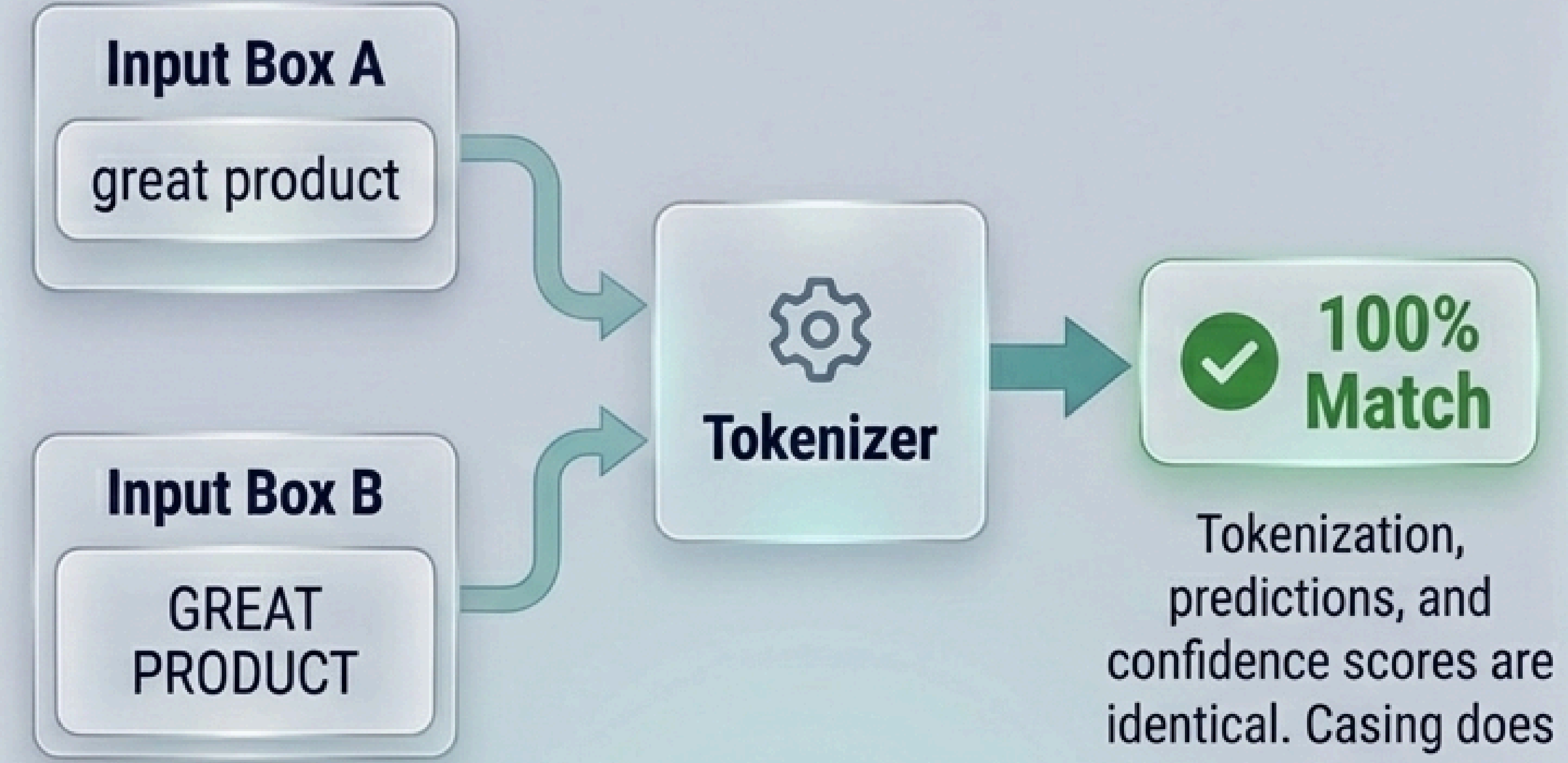
Model Profile

Model: nlptown/bert-base-multilingual-uncased-sentiment

Architecture: 12 attention heads, 12 hidden layers, 768 hidden size.

Training Data: Product reviews in 6 languages (EN, NL, DE, FR, ES, IT).

Case Sensitivity Diagnostic



Tokenization, predictions, and confidence scores are identical. Casing does not affect accuracy.

Sentiment Prediction: English vs. Azerbaijani

Expected	AZ Pred	EN Pred	AZ Conf	EN Conf
5 stars	5 stars	5 stars	51.7%	82.5%
2 stars	1 star	1 star	29.5%	69.1%
2 stars	3 stars	3 stars	41.0%	52.5%

Key Insight: The model is directionally correct, but confidence drops by 25% on average (76.1% EN vs. 51.1% AZ). Why?

The Morphological Map

Diagnostic Summary

Agglutinative Fragmentation: mBERT's tokenizer shatters complex Turkic morphology into disjointed subwords, destroying morphological coherence and drastically reducing downstream confidence.

Track 2: BiDAF Architecture Anatomy

7 **Output Layer** (Start/End position logits)

6 **Modeling Layer** (2-layer BiLSTM)

5 **Attention Flow** (Context-to-Query & Query-to-Context)

4 **Contextual Embedding** (BiLSTM)

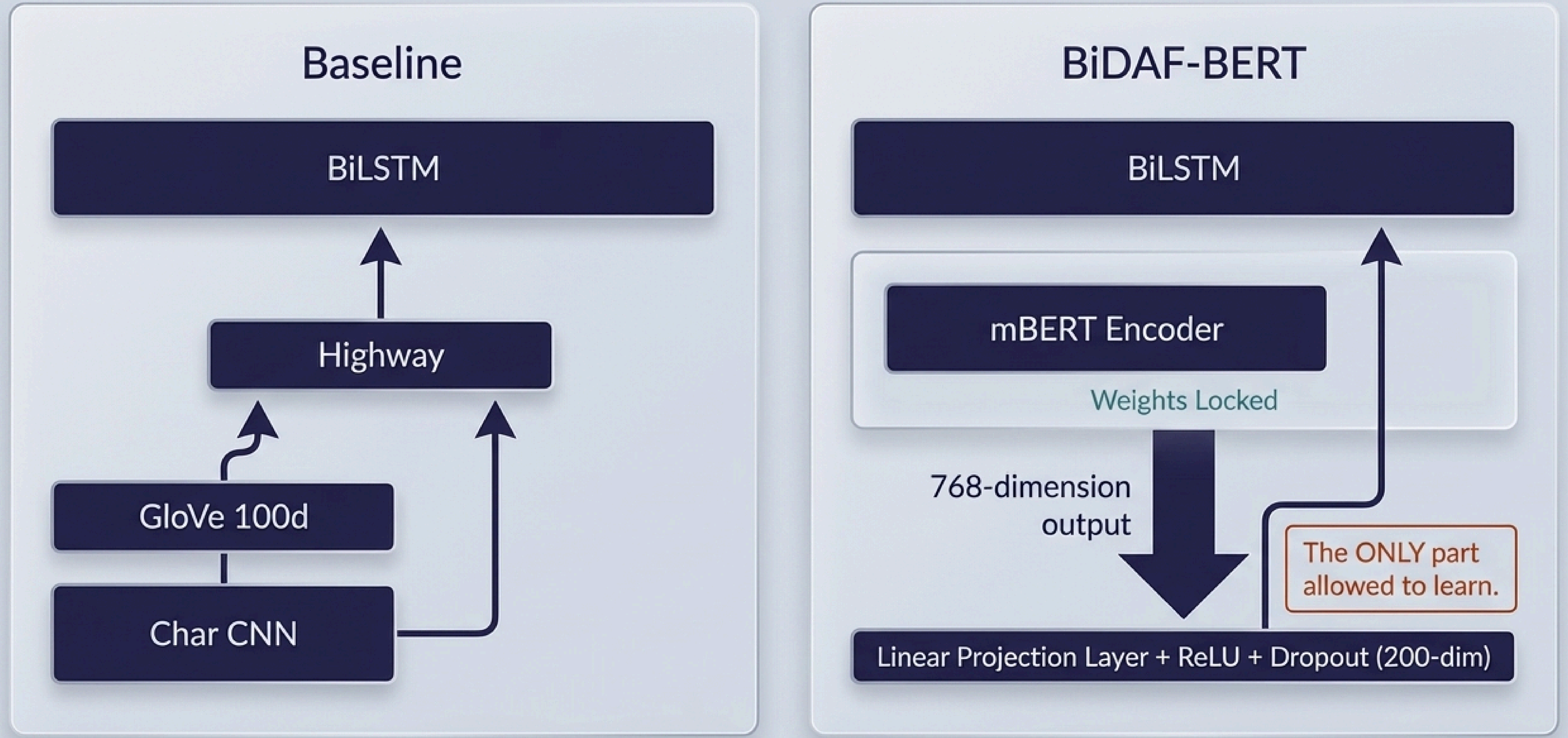
3 **Highway Network** (2-layer)

2 **Word Embedding** (GloVe 6B 100d)

1 **Character Embedding** (CNN, 100 filters)

The Target Zone

The Embedding Swap Experiment



Architecture Showdown Matrix

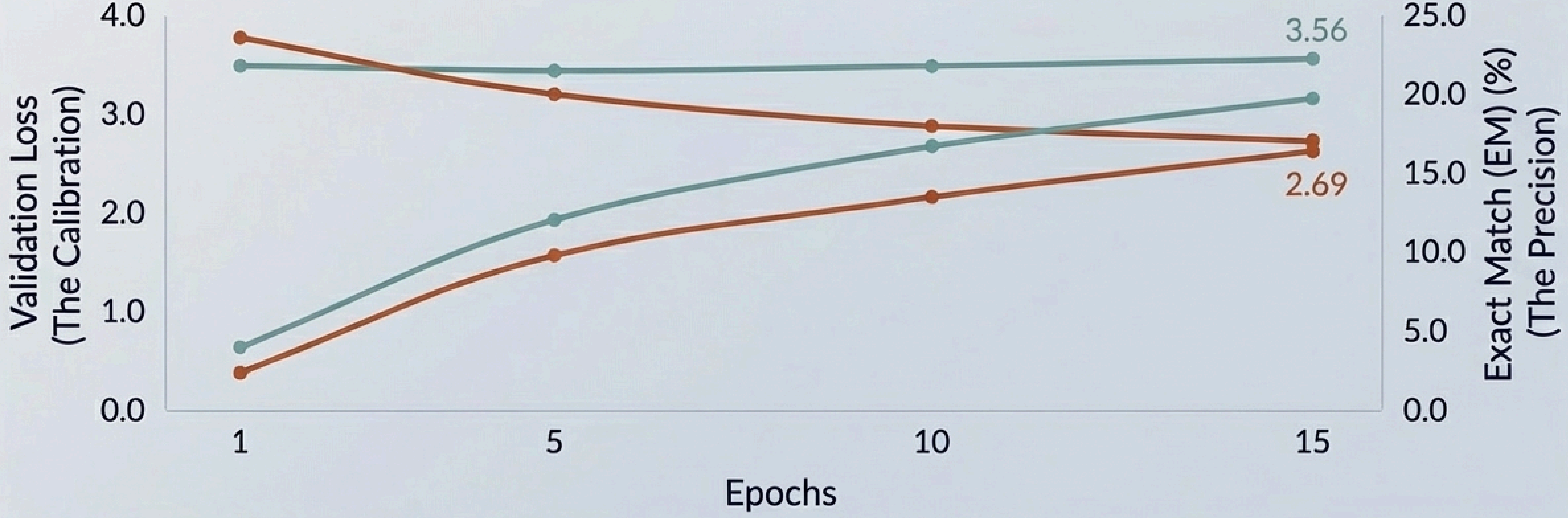
Training Context Badge

SQuAD v1.1 | 5,000 Train / 1,000 Val | 15 Epochs | Adam Opt.

Model	Exact Match (EM)	F1 Score	Parameters
BiDAF (GloVe + Char CNN)	19.70%	30.11%	3.08M (1.6M Trainable)
BiDAF-BERT (Frozen mBERT)	16.40%	24.00%	168.9M (1.6M Trainable)

The Upset: The 3-million parameter model with static embeddings cleanly beats the 168-million parameter transformer by **3.3 EM** and **6.1 F1 points.**

Learning Curve Divergence



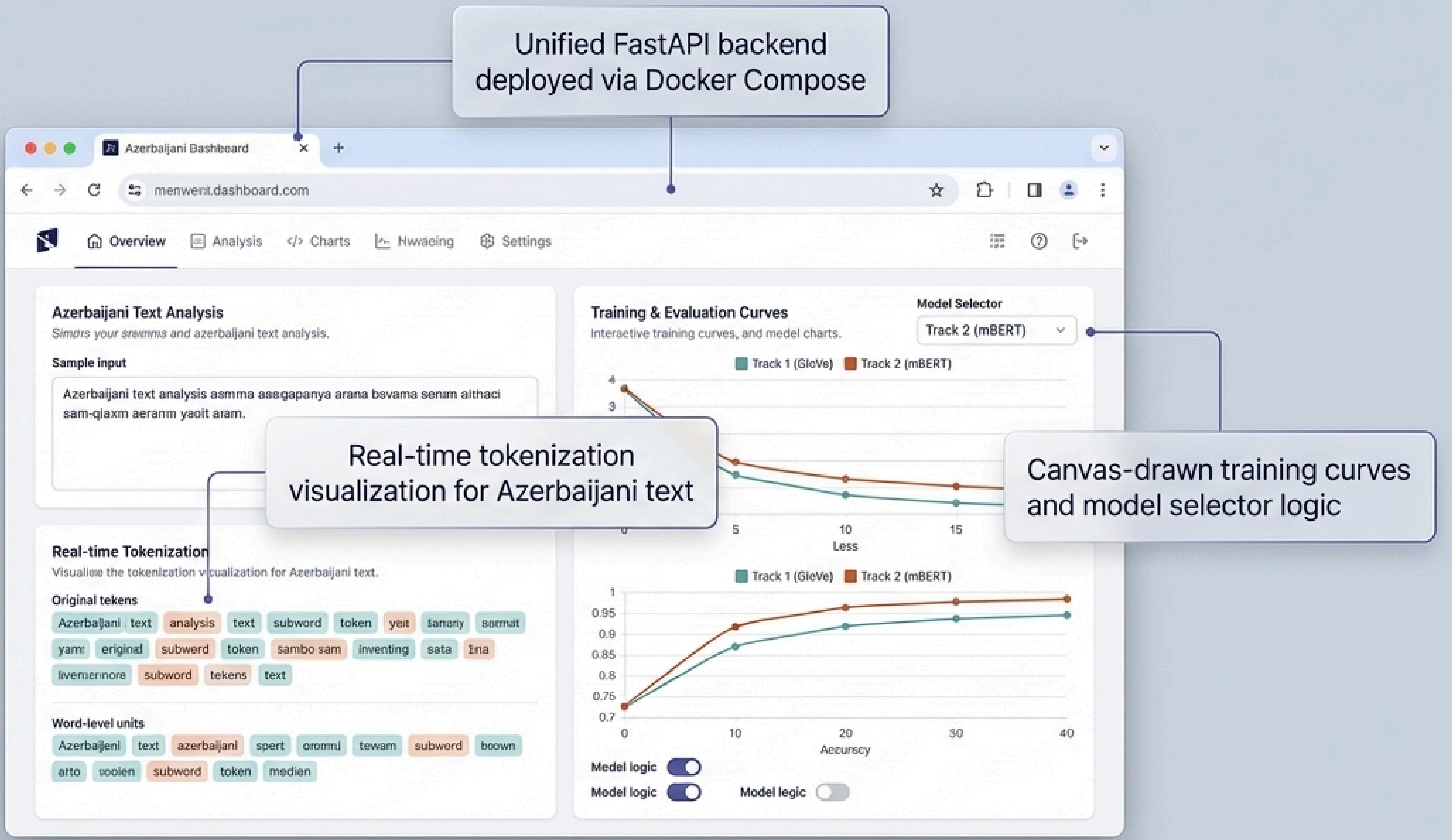
Key Takeaway

The Probability/Precision Disconnect: The frozen BERT variant produces better-calibrated probability distributions (achieving lower loss) but fundamentally lacks the precision required for exact answer span boundaries.

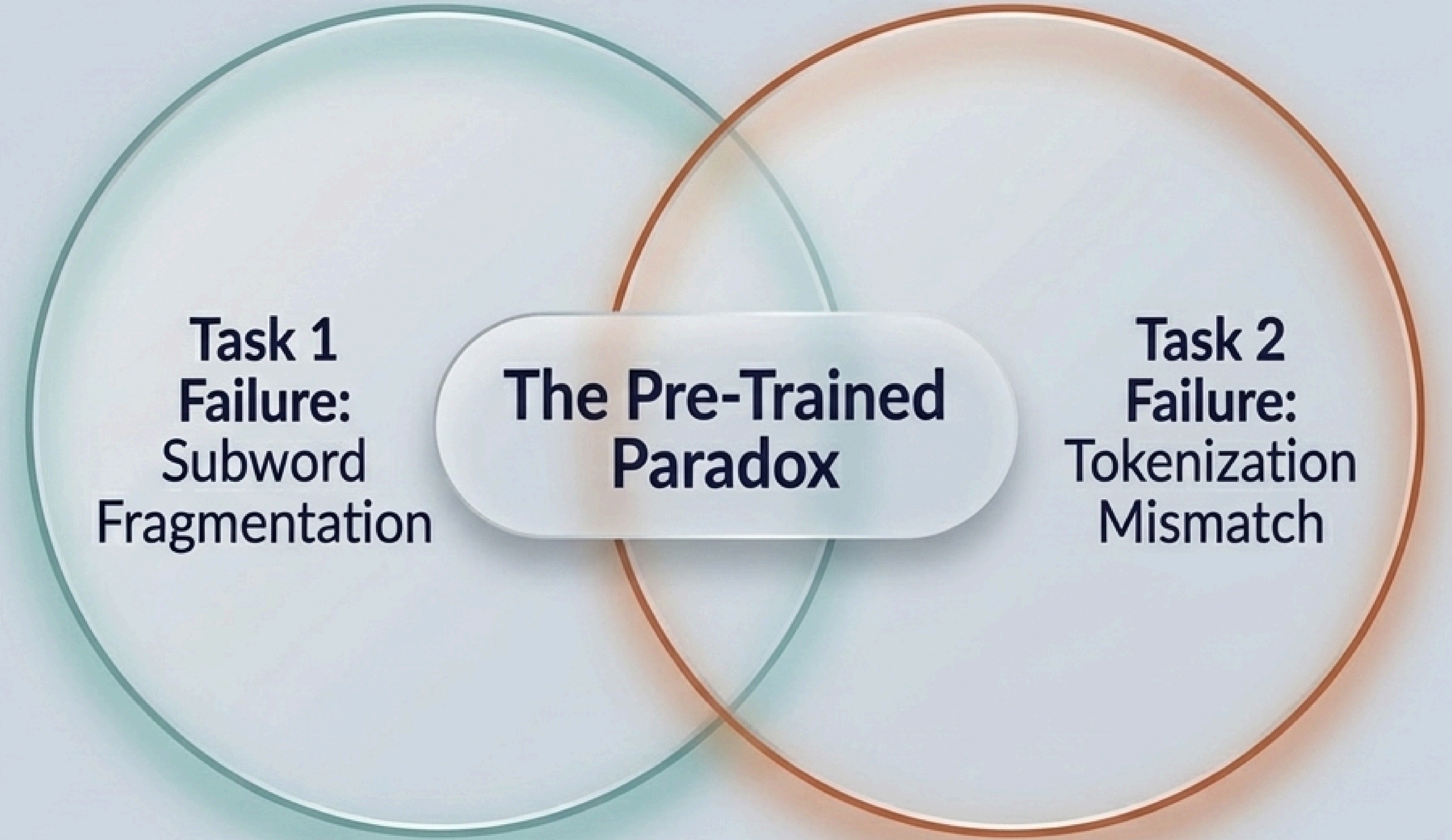
Why GloVe Outperformed mBERT

Diagnostic Report

Factor	Consequence
Frozen Encoder + Small Data (5k)	The 768 to 200 projection layer is the only learned mapping; limited data fails to bridge the gap.
Tokenization Mismatch	BERT uses WordPiece subwords; BiDAF expects word-level tokens. Attention mechanisms dilute the signal.
Small Training Set	Projection layer and downstream components cannot learn an effective mapping from BERT space.
Multilingual Dilution	104-language capacity makes English representations inherently weaker than a dedicated EN-BERT.
Lost Character Features	Removing the Baseline's Character CNN destroys the ability to capture vital morphological patterns.



Extra Task: Engineering rigor extending beyond notebooks into interactive, production-ready diagnostic tools.



Raw embedding quality cannot overcome fundamental morphological or architectural incompatibility. In both tasks, mBERT’s subword tokenition actively fought against the specific needs of the problem (agglutinative syntax in Task 1, word-level span attention in Task 2).

Conclusions & Forward Path

For Agglutinative Languages

mBERT is insufficient for production sentiment analysis on languages like Azerbaijani without dedicated fine-tuning or a Turkic-specific model.

For Architecture Integration

Do not plug-and-play. Architecture-embedding compatibility, training data volume, and fine-tuning strategy matter far more than raw embedding size.

Future Work Roadmap

1. Train on the full 100k+ SQuAD dataset.
2. Swap mBERT for an English-specific bert-base-uncased.
3. Unfreeze the encoder for end-to-end BiDAF-BERT fine-tuning.